Life-Centred Design of a Packaging Value Cycle

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Abstract
Packaging is omnipresent and has evolved through times reflecting the changes in society and in our lifestyles. The modern packaging industry gradually shifted its focus from a product-centred to a human-centred approach and is now starting to embrace a new era in which packaging is considered from a broader perspective — a so-called life-centred design perspective. The reasons behind the constant change are many, from market demands, rapid technological development, rapid population growth, to political steering due to resource and other societal constraints. A life-centred design of packaging value cycles is one that will take into account the well-being of and interaction between all ecosystems. Answers will be discovered through an inclusive exploratory journey of how it should be done.

Keywords
Packaging value cycle
Life-centred design
Sustainability
Resilience
Circular economy

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Change is the only constant in life
— Heraclitus, Greek philosopher

Introduction

Packaging has been present in our lives since ancient times, reflecting and adapting to changes in society in general and in our lifestyles in particular. It fulfils multiple functions, takes different shapes, utilises various raw materials and is increasingly tailor-made to specific uses. In modern times, packaging is still constantly evolving but at a faster pace due to enablers such as globalisation and digitalization. At the same time, societal challenges such as climate change, resource efficiency, littering and resilient society are becoming acute, being ever more regulated on international and national levels. These challenges influence the development of packaging production and consumption.

If we take food as an example, consumers now have the choice between drive-through, home delivery, take-out, cooking at home, and eating out. The model of traditional meals, eaten together, is fading — to be replaced by meals or snacks eaten alone, with each member of the household watching their screen of choice. Today, many teens eat in their bedrooms, often lying on their beds. This highlights the trend of a permanent evolution in lifestyles and consumption (Dairou, Lageat & Delarue, J., 2022). In the political arena, EU legislation such as the Single Use Plastics Directive (EC, 2019), Packaging and Packaging Waste Directive (EC, 2018), regulations on product safety for packaging materials intended for food contact and many more, are becoming more stringent in their requirements for sustainability and circularity of packaging in relation to the environment and the human.

From Product-Centred to Human-Centred Packaging Design

Lockhart (1997) proposed three main criteria for packaging functions, namely a) protection, b) utility, and c) communication. As summarised by Grundey (2010), protection means that a product is shielded from environmental effects during transport and storage. Utility means the packaging gives the consumer an experience of functionality, like spouts on milk cartons or holes in the snap-on shaker inside a spice jar. Finally, the function of communication is about communicating important information to the consumer as well as brand identification, and in some cases advertising and promotion. Extensive research allows us to understand and clarify packaging functions (Brennan et al., 1990; Copley, 2004; Blythe, 2005). Historically, the focus in packaging development was solely on the functional performance of the packaging material with respect to the product packed, i.e. *product-centred design approach*.

In our everyday life, we can observe examples of failure in packaging design for broader value cycles. For instance, one of the reasons why much of plastic packaging is not recyclable today is the industry's previous sole focus on packaging functions and

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making the packaging attractive to the consumer while ignoring material recyclability (KfS, 2018). Here, separation of different types of collected plastics is very difficult, making recycling practically impossible. Furthermore, many materials that are both functional and recyclable are not yet commercially available. This applies, for example, to cheese packaging — for the quality of the cheese to be maintained during its shelf life, the packaging must "breathe", which means that several different types of plastic must be used in one and the same packaging to ensure this packaging property (KfS, 2018).

Meanwhile, packaging design is becoming an important intersection between the product and the user by assisting the latter with information about product use and the benefits the user can expect. A so-called *human-centred design approach* is believed to have been born out of the Stanford University design program in 1958 — when Professor John E. Arnold suggested that engineering design should be human-centred and focus on positive and long-lasting change. The concept has since expanded into numerous fields beyond engineering — including psychology and anthropology — and continues to provide a creative problem-solving solution to help push innovation forward in a practical manner that is grounded in human well-being and user satisfaction (Fast Company, 2022).

In the packaging area, considering the user experience (human-centred design approach) is becoming a key issue for packaging developers. This experience results from the use of the packaging and the product's contents, which together create a complete experience for the user. The experience produced by both must therefore be consistent. The product and its packaging exist together and need to interact if they are to best serve the user. This may seem an obvious aim, but it is often not achieved because many companies have tight deadlines, organisational constraints, and economic pressures to deal with (Dairou, Lageat & Delarue, J., 2022).

One example of human-centred packaging design failure that we can learn from is a packaging-innovation project carried out by Danone, which combined the perspectives of user experience and lowering environmental impact. An innovative and completely recyclable yogurt cup was designed, with lighter packaging material weight. To achieve this goal, the new cup had to be made of both plastic and paper-based components. The packaging developers made it easy for the consumers to tear the two components apart so that each could be discarded in the appropriate recycling bin. The consumer survey results showed that, despite the labels and clear instructions on the packaging, 80% of consumers did not behave in the way that the developers thought they would and later discarded their cups in the wrong bins (Dairou, Lageat & Delarue, J., 2022).

Packaging Dilemmas

We are faced today with a number of dilemmas in packaging value cycles. The value cycles are the opposite conception of "linear economy value chains" where resources are used to "take-make-dispose" of products without further value circulation in terms of variety of value reuse and recycling paths. Dilemmas occur partially due to the complexity of cycles and partially due to focusing only on one

aspect at a time when attempting to solve a problem. This can lead to sub-optimization elsewhere in the value cycle:

- The demand for 100% recyclable food packaging is not compatible with the lack of commercially available functional materials. The risk this creates is to jeopardize food safety, protection, and shelf life. All the while, the environmental impact of food and food waste remains higher than the environmental impact of the packaging it comes in.
- Generally, the consumer is logical and tends to sort waste according to the waste material type such as plastic, paper, metal waste etc. However, in the EU, packaging waste collection under the scope of the Extended Producer Responsibility (EPR) scheme is based on materials application. Separate collection and sorting schemes exist for packaging, electronics, tires etc, rather than for different material types such as paper, plastic, metals. This leads to incorrect sorting of, for example, plastic-based toothbrushes, children's toys, etc., as plastic packaging waste.
- The Packaging and Packaging Waste Directive (PPWD) counteracts value cycle thinking by encouraging packaging recovery by the means of material recycling, incineration or composting. This can be interpreted as considering incineration to be a circular approach, which in practice contradicts the development towards circular material principles. Additionally, the EPR scheme is implemented but lacks control of compliance.
- Designing unnecessary packaging for enhanced user experience.
- Transition to packaging design for reuse offers substantial packaging material savings though it may imply an increase in the costs of logistics and in water usage (during cleaning processes for packaging reconditioning). The net sustainability effect of the transition is still unclear and remains to be realised and measured.

Feber, Lingqvist and Nordigården (2022) describe three disruptive eras in the global and dynamic packaging industry. The era of 2000 – 2009, where the change in substrate (packaging material) occurs in favour of plastic packaging for convenience and low cost. The era of 2009 — 2020, defined by a continued shift in the choice of substrates, expansion into emerging markets and increasingly environmentally-conscious consumers. Finally, the era of sustainability and digitalization starting in 2020 and currently ongoing, with its stricter political measures to foster sustainability and a circular economy. In the EU, this era also includes increased e-commerce and capital allocations based on the sustainability performance of businesses.

Towards Life-Centred Design of Packaging

In the past few years, the design community has begun to question this explicit prioritisation of people and their needs and desires. We can no longer afford the damage wreaked upon the global systems essential to human well-being. Scholars argue that it is necessary to shift to a new design paradigm, creating products with an

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increased focus on human-centred design, that explicitly consider global well-being within the design process (Nardi, 2019). This shift is urgently needed as global well-being is intrinsically linked to the health of global systems, and technology use has had an environmental impact on those systems (Praskievicz, 2021).

The impact of the design practice on the environment has long been recognised in other design sectors (Borthwick, Tomitsch, & Gaughwin, 2022). For example, Papanek (1972) wrote, "There are professions more harmful than industrial design, but only a very few of them". The industrial design sector has responded to these concerns, resulting in various approaches and reference terms, such as eco-design, sustainable product-service systems and the circular economy (Bhamra & Hernandez, 2021). It is becoming evident that we must seek an approach that goes beyond human needs. We need to rethink the way we live and create value in connection with the larger systems that surround us, including nature.

In this sense, the design philosophy called *life-centred design approach* goes one step further. It is understood as a collaborative, inclusive, holistic design approach that expands human-centred design to also consider its sustainable, environmental and social implications (Lutz, 2022). Inglold (2013) believes that part of design's potential is its capacity to speculate and restore environments for life. Tironi, Albornoz and Chilet (2022) further evolve that notion, generating narratives and actions to make us aware of and responsible for the interdependencies that constitute us as earthly beings. One of the challenges of planet-oriented (life-centred) design is precisely to promote modes through which more-than-human agencies, bodies and corporealities that have been ignored by the modern regime can be incorporated into design processes. It is not simply a question of establishing protocols of careful coexistence with these agencies. Rather, it involves developing designs that let us reconnect with the earthly condition, recognizing how these same agencies and bodies can redesign the spaces that we inhabit, generating forms of cohabitation based on mutual care.

Applied to the packaging value cycle, the packaging system is interpreted here as a complex phenomenon with a high degree of economic and cultural cross-sectorality, making design a mediator (Celaschi, 2008) between these processes, needs and actors. The expressions of such a design can lead to new regenerative systems, regrowth of nature, new relationships, reinvented products, manifestos, etc (Spoelstra, 2022).

From a human perspective, according to the Accenture *Survey* (2022a), people are facing a world that feels out of control. At a time when economic, social, environmental and political upheaval is turning almost everything upside down, people are finding themselves in a multi-directional tug-of-war. Faced with the pressure of all these external forces at once, their decisions boil down to trade-offs between what they want, what they need and what options are available. The results can seem contradictory. People are prioritising themselves, but want to effect change for others. They want to follow their values, but not at the expense of value. They're taking matters into their own hands, but also want companies to hold their hand (Accenture, 2022a). These kinds of inconsistencies might not be new, but they're increasingly considered normal — and even good

(Heracleous & Robson, 2020). In fact, the latest Accenture research reveals that up to 69% of consumers globally who admit to behaving inconsistently think that paradoxical behaviours are both human and acceptable (Accenture, 2022b). To stay relevant, businesses must move past customer-centred models and embrace a life-centric view that views people more fully (Accenture, 2022a).

In the area of corporate governance, where strategic decisions are being made about the future of organisations, a growing number of leaders are making statements and manifestos about sustained value creation. As an example, Klaus Schwab (2020), the founder of the World Economic Forum, published the *Davos manifesto* stating that the purpose of companies is more than to exist as economic units generating wealth by fulfilling human and societal aspirations. In the Fourth Industrial Revolution, company performance must be measured not only by the return to shareholders, but also by how the company achieves its environmental, social and good governance objectives. Executive remuneration should reflect stakeholder responsibility. Another example is the initiative of the RISE Research Institutes of Sweden and the Swedish packaging industry association Packforsk in collaboration with GROW, which calls for a paradigm shift in the packaging industry towards truly circular packaging materials and value cycle-based business models (RISE, 2020). Furthermore, the Global Commitment 2022 (Ellen Macarthur Foundation, 2022) unites businesses and governments to commit to change how we produce, use, and reuse plastic, including companies representing 20% of all plastic packaging produced globally.

Holistically, the life-centred design of a packaging value cycle can be visualised conceptually within a system-thinking perspective, giving it the ability to interconnect within and between ecosystems while creating sustainable value in economic, social and ecological terms (see Fig. 1). It triggers new ways of innovating, with value cycles designed in an explorative and iterative manner.

The varying design practices used by different life-centred approaches draw from a mix of circular design, systems thinking, biomimicry, futures studies, and others (Lutz, 2022). Here, the circular economy is based on three principles, driven by design: eliminate waste and pollution, circulate products and materials (at their highest value) and regenerate nature (Ellen MacArthur Foundation, 2023). While there is no single solution, one may consider designing packaging that becomes an integrated part of the product it packs, thereby eliminating the need for packaging waste management. One such example is the cellulose fibre-based Billerud D-Sack® cement packaging that “disappears” or dissolves when mixed with the cement (product), water and other components for concrete production (Billerud, 2016). Another may consider designing packaging for technical (e. g. material recycling) or biological (e. g. biogas production) systems to create circular packaging value cycles. All in all, the packaging value journey should be explored through systems thinking in order to maximise the potential of sustainability and minimise the potential of risks.

In this journey, important questions arise: where is the packaging value created and for whom? Is the value created in the production

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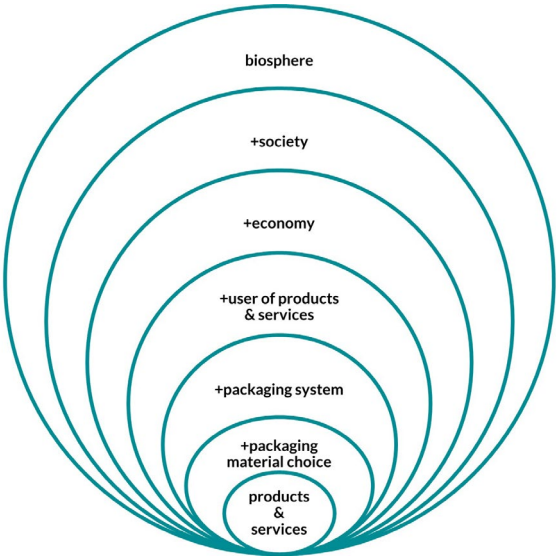


Fig. 1
Conceptual visualisation of systems thinking as the ability to interconnect within and between systems such as ecosystems, including economic (value creation through products and services), societal and ecological environments, elaborated through the lens of a packaging value cycle.

processes or is it created by the user, and in this case who is the user (e. g. a direct user or also nature as user or both)? How do we define and measure environmental, economic and social values inclusively? Thus, in the packaging area, the focus clearly evolves from packaging production in terms of optimization of packaging material performance (technical properties) to packaging as a service where value is created for broader, multi-dimensional ecosystems. Under consideration are the inclusive perspective of a product to be packaged, consumer needs and behaviours, as well as possibilities and limitations of the larger economic, social and environmental systems that surround us.

The Open Debate Section in a Nutshell

Following the Fig. 1 life-centred design approach, the article by Béland and Granberg in this Open Debate section is dedicated to the packaging material creation process where multiple potential material properties are discovered and matched with multi-dimensional performance requirements of the products they shape and the environment. By introducing material demonstrators into the material innovation process, the article pinpoints nine roles that demonstrators can have to help reduce the lab-to-market gap. One of the important insights in the article is related to an early interaction between researchers and designers through the tangible and inspiring media of material demonstrators. This creates a deeper mutual understanding of sensory material qualities and the Research, Development and Innovation (R&D&I) process and thus accelerates the innovation process. The article presents a practitioner's point of view on how coupling technology-driven results with user(market)-driven needs can be facilitated by a multidisciplinary approach.

Dominic and Olsmats dive into the life-centred packaging design approach through the lens of a four-dimensional concurrent engineering (4DCE) approach for the development of product, logistics process, packaging and supply chain. The case study from the fresh food industry in which a holistic packaging development concept was used pinpoints the complexity and interconnections within and between the different systems in a packaging life cycle.

The research by Crisigiovanni and Agante unleashes the user perspective. Enablers of sustainable products and services, including packaging, such as digital platforms with information about the product life cycle, choices of biobased and circular packaging materials and guidelines for municipalities were detailed in order to identify user barriers and incentives for change and find common behavioural patterns. One of the research findings strongly involved design, which may determine 80% of the environmental impact that a product or service generates in its life cycle.

Furthermore, Huerta provides a deep insight into designing and developing packaging with environmental responsibility. By applying a hybrid approach of prescriptive (such as environmentally responsible design) and descriptive (such as Life Cycle Assessment (LCA)) methods in packaging design, the ultimate objective is to minimise the environmental impact of products and services, thus encouraging regenerative ecosystems.

Giardina elaborates on the primary need to design in a sustainable and circular way. In the contemporary "polycrisis", she emphasises the role of design and the emergent need for Transitional Industrial Designers, with their approaches, strategies and tools, in order to succeed in both including and simplifying all the new variables of contextual complexity within a project.

Pletto expands the topic to the global network of institutes, research centres and observatories - key players in the development and innovation of packaging systems. The analysis identifies regional concentrations of areas of expertise, supported by six selected case studies that illustrate different experimental approaches to packaging research.

Indeed, today's environmental, social and economic sustainability challenges are so complex that they can no longer be solved by scientific thinking or creative thinking alone, both need to be combined into an intertwined and inclusive approach, i.e. life-centred design. In exploring the life-centred design of packaging value cycles, it is important to consider that sustainable packaging is as sustainable as the system you are choosing to design it for. With both scientific and creative courage, we can together start the journey of designing well-being and healthy interaction within and between all ecosystems thereby creating a positive impact from packaging value cycles.

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She has 15 years of experience in sustainability and circularity strategies for materials in packaging value cycles. At RISE, which is an industrial research institute, she supportssmall & medium size companies in their transition to sustainable packaging by taking into account a holistic approach to sustainability, i.e., the environmental, economic and social sustainability implications.

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Doi: 10.30682/diid7923a

References

Accenture. (2022a). *The human paradox. From consumer centricity to life centricity*. https://www.accenture.com/_acnmedia/PDF-180/Accenture-Human-Paradox.pdf

Accenture. (2022b). *Accenture Research, global consumer survey*. https://www.accenture.com/content/dam/accenture/final/industry/banking/document/Accenture-Banking-Consumer-Study.pdf

Billerud. (2016). *Tänk om förpackningen bara kunde försvinna?* https://www.billerud.se/branscher/industri/industriella-pasar/billerudkorsnas-d-sack

Bhamra, T., & Hernandez, R. J. (2021). Thirty years of design for sustainability: An evolution of research, policy and practice. *Design Science*, 7. https://doi.org/10.1017/dsj.2021.2.

Blythe, J. (2005). *Essentials of Marketing*. Prentice Hall & Financial Times, Pearson Education.

Borthwick, M., Tomitsch, M., & Gaughwin, M. (2022). From human-centred to life-centred design: Considering environmental and ethical concerns in the design of interactive products. *Journal of Responsible Technology*, 10:100032, 1-10.

Brennan, J. G., Butters, J. R., Cowell, N. D., & Lilley, A. E. (1990). *Food Engineering Operations*. Elsevier App. Sci.

Celaschi, F. (2008). Il design come mediatore tra bisogni. Il Design come mediatore tra saperi. In C. Germak (Ed.), *L'uomo al centro del progetto* (pp. 19-31). Allemandi.

Copley, P. (2004). *Marketing Communications Management: concepts and theories, cases and practices*. Elsevier.

Dairou, V., Lageat, T., & Delarue, J. (2022). 8 - The packaging experience shift: Moving from product-centric to life-centric design. In A.-M. Pensé-Lhéritier, I. Bacle, & J. Delarue, *Woodhead Publishing Series in Food Science, Technology and Nutrition* (pp. 153-168). Nonfood Sensory Practices, Woodhead Publishing.

EC. (2018). *European Parliament and Council Directive 94/62/EC of 20 December 1994 on packaging and packaging waste*. https://eur-lex.europa.eu/legal-content/en/TXT/?uri=CELEX:31994L0062

EC. (2019). *European Union Directive (EU) 2019/904 of the European Parliament and of the Council of 5 June 2019 on the reduction of the impact of certain plastic products on the environment*. https://eur-lex.europa.eu/eli/dir/2019/904/oj

Ellen Macarthur Foundation. (2022). *The Global Commitment 2022 in collaboration with the UN Environment Programme*. https://www.unep.org/resources/report/global-commitment-2022

Ellen MacArthur Foundation. (2023). *Circular economy introduction. What is a circular economy?* https://ellenmacarthurfoundation.org/circulate-products-and-materials

Fast Company. (2022). *What is human-centered design? Here is why it is so important*. https://www.fastcompany.com/90772846/human-centered-design

Grundey, D. (2010). Functionality of Product Packaging: Surveying Consumers' Attitude Towards Selected Cosmetic Brands. *Economics & Sociology*, *3*(1), 87-103.

Heracleous, L., & Robson, D. (2020). Why the 'paradox mindset' is the key to success. *BBC*, November 11, 2020.

Ingold, T. (2013). *Making: Anthropology, Archeology, Art and Architecture*. Routledge.

KfS. (2018). *Konsumentföreningen i Stockholm, Plastförpackningar med konsumenterna* för *en hållbar relation till plast*. KFS rapport.

Lockhart, H. (1997). A paradigm for packaging. *Packaging Technology and Science*, 10, 237-252.

Lutz, D. (2022). *Life-centred design*. UX Collective.

Feber, D., Lingqvist, O., & Nordigården, D. (2022). *2022 and beyond for the packaging industry's CEOs: The priorities for resilience*. McKinsey & Company.

Nardi, B. (2019). Design in the age of climate change. *She Ji: The Journal of Design, Economics, and Innovation*, *5*(1), 5-14.

Papanek, V. (1972). *Design for the Real World*. Pantheon.

Praskievicz., S. (2021). How the environment became global. *Anthropocene*, 35. https:// doi.org/10.1016/j.ancene.2021.100305

RISE. RISE Research Institutes of Sweden, Swedish packaging industry association Packforsk and GROW. (2020). *Rethinking the Packaging System – Innovation Manifesto*. https://www.ri.se/en/our-stories/time-for-a-paradigm-shift-in-the-packaging-industry

Schwab, K. (2020). *Davos manifesto 2020: The universal purpose of a company in the fourth industrial revolution*. World Economic Forum.

Spoelstra, J. (2022). *What is Life Centered Design?* Unbeated studio.

Tironi, M., Albornoz, C., & Chilet, M. (2022). Problematizing Human-Centred Design: Notes on Planet-Oriented Design, *Diid — Disegno Industriale Industrial Design*, 77, 38-49. 10.30682/diid7722c

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Doi: 10.30682/diid7923a